Fiscal Year:	FY 2007	**Task Last Updated:**	FY 09/27/2006
PI Name:	Freed, Lisa E M.D., Ph.D.		
Project Title:	Microgravity Tissue Engineering		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Cell Culture		
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Zip Code:	02139-4037	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Space Biology 01-OBPR-08
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No. of Post Docs:	4	**No. of PhD Degrees:**	0
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Changed Division and Discipline/Program to FSB as of FY2006, per program changes at that time, per J. Cohen/HQ and JSC/J. Dardano and previous information fr D. Risin/JSC (jvp 6/2009) NOTE: Received NCE to 11/30/2008 (from 11/30/2007) per P. Denkins/JSC (7/07) NOTE: Funding reinstated to 11/30/2007, per J. Cohen (HQ, 8/2006)		
Key Personnel Changes/Previous PI:	Prof. Laura Niklason is no longer working on this project since she has left Duke University.		
COI Name (Institution):	Vunjak-Novakovic, Gordana (Columbia University and M.I.T.) Rupnick, Maria (Harvard Medical School) Guilak, Farshid (Duke University Medical Center) Padera, Robert (Harvard Medical School)		
Grant/Contract No.:	NNJ04HC72G		
Performance Goal No.:	1.21		
Performance Goal Text:	Sustain a leading biotechnology research program that will assure continued scientific and technical leadership.		

Task Description:

Task Description: Our research project involves cell- and tissue-levels studies of 3D tissue constructs using models of relevance to tissue loss in astronauts in microgravity and skeletal and myocardial tissue repair in human patients on earth. We selected to study skeletal and cardiac tissues because: (a) spaceflight adversely affects these tissue types, (b) improved treatments are needed for millions of patients who suffer from progressive skeletal and cardiac diseases, (c) these two distinctly different tissue types can provide paradigms for studies of individual and interactive effects of biophysical stimuli, and (d) we and others have already demonstrated the feasibility of engineering skeletal and cardiac-like tissue, and thus can capitalize on recent advances to promote groundbreaking research and technology transfer.
We previously designed an in vitro system based on 3D cell-biomaterial constructs and advanced bioreactor systems, and demonstrated that bioreactor-grown constructs were larger and functionally superior to controls grown in petri dishes, as assessed by quantitative structural, molecular, and functional assays. The current limitations of our tissue engineering research are as follows: (i) functional cartilage constructs based on human cells have not yet been made, and incomplete integration of cartilage constructs with adjacent host tissues may limit their long-term mechanical properties, (ii) cardiac tissue constructs have only been made in very thin layers and from non-human cells that are present at low number and low viability, and (iii) spaceflight tissue culture studies are limited by the lack of proper controls necessary to derive underlying mechanisms. We are addressing these limitations by: (a) integrated studies in cell and molecular biology, mass and momentum transport, design and operation of bioreactors including microgravity analog systems, tissue physiology and biomechanics, and implant biocompatibility and remodeling, (b) experiments interpreted with the aid of mathematical models, (c) complementary in vitro and in vivo studies, and (d) functional evaluation at various hierarchical scales, i.e. molecular, cell, and tissue.

The project is organized as two Specific Aims: (1) functional skeletal tissue engineering, and (2) functional cardiac tissue engineering. The global hypothesis is that biophysical stimuli that are physiologic in nature, magnitude and regime of application can direct cells to regenerate functional tissues in vitro. We can test this hypothesis in controlled studies of cells cultured on 3D scaffolds in bioreactors and in the presence of biophysical stimuli. The work is expected to help define how biophysical cues regulate skeletal and cardiac tissue development and function, and to yield engineered skeletal and cardiac tissues with superior functionality to that previously reported in the literature.

Task Objective: Our studies of skeletal and cardiac tissue constructs involve the cultivation of cells on 3D scaffolds in bioreactors, including microgravity analog systems, and in the presence of biophysical stimuli. Critical research areas include cell sourcing (cell expansion, stem cell differentiation), development of novel biomaterial scaffolds, cell seeding of scaffolds, construct cultivation in vitro for controlled studies of biophysical stimuli (e.g. hydrodynamic shear, cyclic stretch, growth factors, pharmaceutical agents), and construct implantation in vivo for studies of construct survival, integration, and capacity for defect repair. The constructs are assessed with respect to structure, composition, and mechanical function to evaluate individual and interactive effects of experimental variables. Key equipment includes facilities for working with cells, explanted tissues, and experimental animals, polymer processing equipment, bioreactor systems, biomechanical and electrophysiological test systems, and analytical instruments for histology, biochemistry, molecular biology, microscopy and image analysis. Construct properties are correlated with in vitro culture conditions and the outcome of in vivo studies, in order to provide feedback necessary to tailor model system parameters and obtain the best overall result.

Task Significance: Our research is relevant to the Bioastronautics Roadmap (BR) questions #2) impaired fracture healing, #3) injury to joints, #5) occurrence of serious cardiac dysrhythmia, #6) diminished cardiac function, #19) pharmacology of space medicine, and possibly also #21) rehabilitation on Mars < http://bioastroroadmap.nasa.gov/ ;. Our combined use of 3D tissue constructs and advanced bioreactor systems provides a unique model system for controlled in vitro studies at the cell and tissue-levels that may be relevant to treating skeletal and myocardial tissue loss and dysfunctions in astronauts. Biophysical stimuli play a critical role in tissue growth, maintenance, and remodeling. During human spaceflight, altered biophysical stimuli and the resulting systemic adaptations can cause fluid and electrolyte imbalance, altered neurohormonal regulation, diminished tissue remodeling, and loss of bone and muscle mass. Exercise regimens and pharmaceutical interventions have been proposed to try to maintain skeletal and cardiac tissues in-flight, and to try to rehabilitate deconditioned skeletal and cardiac tissues post-flight. However, the ability to determine the underlying mechanism(s) or assess the potential risks/benefits of currently used countermeasures is limited by a lack of experimental models in which to study development and remodeling of skeletal and cardiac tissues. Our controlled studies, in high fidelity in vitro models, of how 3D constructs respond to biophysical stimuli are expected to help identify how the microgravity environment of space may compromise the ability of skeletal and cardiac tissues to respond to and/or withstand biophysical stimuli. Improved understanding is expected to aid in the development of better countermeasures.

Our tissue engineering research has implications not only for astronauts but also for humans on earth, e.g. for obesity related degenerative joint diseases and hypertensive cardiomyopathy. Skeletal and cardiac tissues perform structural and mechanical functions vital for health and survival. Incidences of osteoarthritis and heart failure are increasing and, in the absence of curative interventions, have resulted in substantial human suffering and medical expense. Over forty million Americans suffer from arthritis, which is characterized by the degeneration of cartilage and results in chronic pain and progressive disability. Once damaged, cartilage has a limited capacity for self-repair, and no currently available therapy can predictably restore a functional articular surface. Over five million Americans suffer from heart failure, which is characterized by the degeneration of cardiac muscle and is a leading cause of morbidity and mortality in developed countries. The only currently available definitive treatment for end stage heart failure is cardiac transplantation, but limited organ availability leads to long waiting periods that are often not survivable. Tissue engineering is emerging as a significant clinical option to address tissue and organ failure by implanting a 3D tissue construct that can provide immediate functionality and the capacity to grow and integrate with surrounding host tissues. However, although cell-based procedures for articular cartilage repair procedures are already in widespread clinical use, critics maintain that the field is principally empirical in nature, and the results unsatisfactory with respect to integrative repair and durability. Significant challenges also remain to be solved before cell-based procedures for cardiac repair can be used clinically. Our research is thus expected to be of critical importance for the human exploration of space and also to benefit life on Earth.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

To date, the present NASA-funded research has been highly productive, yielding more than ninety publications in leading peer-reviewed scientific journals including the one-and-only long duration microgravity tissue engineering experiment carried out aboard STS-79, Mir Space Station and STS-81, and published by Freed et al., Proceedings of the National Academy of Sciences USA. The present NASA project has also resulted in more than 120 abstracts and five patent applications. To date, the present NASA-funded research has trained sixteen postdoctors, nine graduate students including six who earned PhD degrees, and numerous undergraduates. The co-investigators use research supported by this NASA project as examples in the courses that we teach. For example, L.E. Freed teaches in graduate-level courses in the Harvard-MIT Division of Health Sciences and Technology, e.g. "Biomaterials and Tissue Engineering for Medical Devices" (HST-521), and "Principles and Practice of Tissue Engineering" (HST-535), and F. Guilak teaches in courses at Duke University, e.g. "Stem Cell Biology" (MCB/CBI-208), "Introduction to Biologically Inspired Materials" (MEMS-265/BME-265), the Rheumatology Basic Science Series, the Osteoarthritis Series, and the Center for Biomolecular and Tissue Engineering. Other professional activities of the co-investigators include chairing and organizing scientific conferences, and serving on scientific research panels and scientific and editorial advisory boards. Also, NASA has used the data and experience gained in our ground and flight studies to develop two bioreactor systems: (i) Advanced Rotating Bioreactor System and (ii) Cell Culture Unit. Our ongoing work has thus resulted in new insights into generating functional cartilage and cardiac tissues, which have had a major impact on America, and have benefited NASA and life on Earth.
Innovative Technologies Developed: MIT Technology Disclosure No. 12301, filed July 20, 2006, "Oscillating Cell Culture Bioreactor". Freed, L.E., Langer, R., Moretti, M.

Patent Application No. 10/872,577, filed June 21, 2004, "Application of Electrical Stimulation for Functional Tissue Engineering in vitro and in vivo". Freed, L.E., Langer, R., Park, H., Radisic, M., Vunjak-Novakovic, G. (patent application also filed in Europe).

Patent Application No 10/457,248, filed June 9, 2003, "Effect of Bone Morphogenetic Proteins on Engineered Cartilage" Blunk, T., Courter, D.L., Freed, L.E., Gooch, K.J., Sieminski, A.I., Vunjak-Novakovic, G.

Patent Application No 09/809,456, filed March 15, 2001, "Tissue Engineering Enhanced by the Transfer of Growth Factor Gene", Freed, L.E., Langer, R., Madry, H., Trippel, S. Vunjak-Novakovic, G.

US Patent No. 6,582,960, issued June 24, 2003 "Use of Fibroblast Growth Factor-2 for Expansion of Chondrocytes and Tissue Engineering", Freed, L.E., Langer, R., Martin, I., Vunjak-Novakovic, G. (patent applications also filed in Europe, Canada and Japan).

Task Progress:

Aim 1 (selected progress). Our skeletal tissue engineering studies involve chondrocytes or chondroprogenitor cells, 3D scaffolds, bioreactors, and biophysical signals. In one study we tested whether large cartilaginous and bone-like constructs could be selectively generated by using human bone marrow stromal cells in conjunction with a porogen-leached silk scaffold and rotating bioreactors, and supplementing the culture media with either chondrogenic or osteogenic differentiation factors.(Marolt et al., Biomaterials (in press). The stem cells were harvested, expanded and characterized with respect to their differentiation potential and population distribution. Chondrogenic medium yielded constructs with higher amounts of DNA and glycosaminoglycans than control medium. In contrast, osteogenic medium yielded constructs with higher alkaline phosphatase activity and calcium content than control medium. Chondrogenic medium yielded constructs that were weakly positive for glycosaminoglycans by contrast-enhanced MRI and alcian blue stain, whereas osteogenic medium yielded constructs that were highly mineralized by microCT and von Kossa stain. Engineered bone constructs were large (8 mm diameter x 2 mm thick disks) and resembled trabecular bone with respect to structure and mineralized tissue volume fraction. The data show that rotating bioreactors provide a good model system for generating large bone-like constructs from human stem cells.
Biomaterial scaffolds provide a critical means of controlling engineered tissue architecture and mechanical properties (Freed et al., Tissue Eng. in press). Many previous cartilage tissue engineering studies have employed scaffolds in which architectural features are present in isotropic spatial orientation, such as non-woven mesh of polyglycolic acid (Freed et al., Nature Biotechnology 1994). However, biomechanical properties and low mechanical strength of isotropic scaffolds may not be suitable for the repair of anisotropic, load-bearing tissues in some clinical applications. In an attempt to recreate the complex, multiphasic mechanical properties of native articular cartilage, we used computer-controlled weaving technologies to recreate anisotropy, viscoelasticity, and tension-compression non-linearity of the extracellular matrix of cartilage (Moutos et al., submitted). A novel microscale 3D weaving technique was used to weave polyglycolic acid yarns into an orthotropic, porous textile that was then infiltrated with a hydrogel. The scaffold displayed significant tension-compression nonlinearity, with approximately three orders of magnitude difference in tensile and compressive moduli, an aggregate modulus of ~0.2 MPa and a tensile modulus of ~300 MPa. Significant anisotropy was observed in the failure stress, failure strain, tangent modulus, and energy-to-failure. These findings show that 3D woven scaffolds can be designed that mimic some of the complex biomechanical properties of native articular cartilage, even in the absence of cells. Moreover, the large number of variables in this design (selection of fiber material, number of fibers up to several hundred, density and packing in all directions) provides a wide range of possible scaffold mechanical properties.

Aim 2 (selected progress). Our cardiac tissue engineering studies involve heart cells, 3D scaffolds, bioreactors, and biophysical signals. In one study (Radisic et al., Tissue Eng, 2006), heart cells were used in conjunction with a porogen-leached polyglycerol sebacate scaffold and bioreactors to generate cardiac tissue constructs. The constructs were cultured with direct perfusion of media that was supplemented or not with perfluorocarbon particles that provided a reservoir of dissolved oxygen. Supplemental perfluorocarbons increased construct amounts of DNA and muscle marker proteins, and yielded cardiac-like tissue better contractile properties than controls cultured in unsupplemented media. A mathematical model of oxygen distribution was developed and oxygen gradients were correlated with construct cell density and viability (Radisic et al., Biotech Bioeng, 2006). In another study (Radisic et al., submitted), sequential seeding and culture of first cardiac mesenchymal cells and then myocyte-enriched heart cells on polyglycerol sebacate scaffolds in bioreactors yielded constructs with significantly higher contractile amplitude than seeding and culturing myocyte-enriched heart cells in a single step. Importantly, sequential seeding and culture yielded constructs that exhibited: (i) similar final fractions of cardiomyocytes, (ii) similar final amounts of cardiac-specific proteins, and (iii) increased collagen deposition, compared to culturing only the myocyte-enriched heart cells. The aforementioned findings were in contrast to our previously held notion that cell sources used to make cardiac tissue constructs should be depleted of fibroblasts by pre-plating lest they overgrow the cardiomyocytes (e.g. Bursac et al., Am J Physiol-H, 1999).

	We also evaluated whether cardiac tissue constructs could be used as a model for controlled in vitro studies of pharmaceutical drugs and regulatory factors. In one study, sponges made of collagen were seeded with cardiomyocytes and cultured in media supplemented or not with three positive inotropic drugs (digoxin, milrinone, or isoproterenol). The drugs affected cell metabolic activity (based on the molar ratio of lactate to glucose in media), construct molecular composition (based on amounts of muscle markers by Western blot), and electrophysiological function (excitation threshold and contractile amplitude)(Cheng et al., CIMIT Abstract, 2006). Another study tested the hypothesis that supplemental growth and regulatory factors could improve the properties of engineered cardiac tissues made by cultivation of cardiomyocytes on porous collagen sponges (Cheng et al. BMES Abstract, 2006). Constructs cultured in media supplemented with insulin, insulin-like growth factor, or platelet-derived growth factor were compared to unsupplemented constructs and native heart tissue with respect to cell viability and metabolism and construct electrophysiologic and molecular properties. Cellular release of lactate dehydrogenase (an indicator of cell death) was significantly reduced by all of the regulatory factors, and programmed cell death (apoptosis) was significantly reduced. Electrical stimulation induced synchronous contraction of constructs at a threshold that was significantly decreased by insulin and insulin-like growth factor-I. Also, construct amounts of a cardiac marker protein (troponin-I), total protein and contractile amplitude were significantly increased by insulin and insulin-like growth factor-I. Together, the findings demonstrate that engineered cardiac tissue constructs provide a high-fidelity model to for molecular-, cell- and tissue-level studies of drugs and regulatory molecules. The constructs thus provide a model for controlled in vitro studies of the effects of complex stimuli that occur in vivo, and are expected to (i) improve our basic understanding of cardiogenesis and (ii) help identify underlying mechanisms that can potentially be exploited in order to improve cardiact repair in vivo.

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Patents	MIT Technology Disclosure No. 12301. filed July 20, 2006. Jul-2006 Freed LE, Langer R, Moretti M. "Oscillating Cell Culture Bioreactor."